Angel Hiram Alvidrez Hernandez

Mechatronics Engineering Student | Robotics & Embedded Systems

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Portfolio: angelalvidrez.github.io

EDUCATION

Universidad Autónoma de Ciudad Juárez (UACJ) *Ciudad Juárez, MX*
B.S. in Mechatronics Engineering *Expected: 2027*
Award: "Estudiante Integral" (Well-Rounded Student Award) for academic & community excellence (2025).

TECHNICAL SKILLS

- **Programming:** C/C++ (Advanced), Python, MATLAB, Assembly.
- **Hardware:** ESP32 (Dual Core), Raspberry Pi, Orange Pi, Arduino.
- **Robotics:** Inverse Kinematics, PID Control, Sensor Fusion (IMU).
- **Protocols:** I2C, UART, SPI, WiFi/Bluetooth (IoT).
- **Electronics:** PCB Design, OpAmps, Power Electronics.
- **Software:** SolidWorks, Fusion 360, Proteus, KiCad.

FEATURED PROJECTS

View full code, demos, and documentation at: angelalvidrez.github.io

Underactuated Biped Robot (5-Bar Mechanism) *2025*

Tech Stack: C++, ESP32, Orange Pi, BNO08x IMU, Inverse Kinematics.

- Designed and built an autonomous bipedal robot utilizing a 5-bar linkage mechanism.
- Implemented **inverse kinematics** to generate cycloidal trajectories for stable walking gaits.
- Developed a stability system using **Sensor Fusion (BNO08x)** for real-time posture correction.
- Master-Slave Architecture: Orange Pi (Trajectory planning) and ESP32 (Servo control).

8" Orion Telescope Control System Upgrade *2025*

Tech Stack: Python, Raspberry Pi, ESP32, Optical Encoders.

- Performed reverse engineering to modernize the control electronics of a commercial telescope.
- Developed a **Python** interface on Raspberry Pi to communicate with **Stellarium** software.
- Designed signal conditioning circuits for optical encoders and DC motor control (H-Bridges).

IoT Smart Irrigation System *2025*

Tech Stack: IoT, Blynk, ESP32, Environmental Sensors.

- Created an IoT ecosystem for efficient water management with remote monitoring via Mobile App (Blynk).
- Integrated soil moisture, DHT11 (temp/humidity), and PIR sensors for security alerts.
- Implemented local data visualization using an LCD screen via I2C protocol.

EXPERIENCE & LEADERSHIP

Institute of Astronomy, UNAM (OAN-SPM) *June - July 2025*
Research Internship - XXXIV Scientific Summer Program

- Collaborated on astronomical research projects and specialized workshops at the National Astronomical Observatory.

Scientific Outreach / Science Communication *Ciudad Juárez, MX*

- Speaker: "Are we alone in the Universe?" - XXX Engineering Week (2024).
- Speaker: "Deaths of the Universe" - XXIX IIT Week.

CERTIFICATIONS & LANGUAGES

- **English Proficiency (TOEFL ITP):** Score 633/677 (Level C1 - Advanced), Oct 2025.
- **Semiconductor Fundamentals** - CENALTEC (90 Hours), Dec 2024.